Code No: 115EH **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2024
OPERATING SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	What are the goals of an operating system?	[2]
b)	What are the features of parallel systems?	[3]
c)	What is the role of medium term scheduler?	[2]
d)	What is critical section problem?	[3]
e)	What is a page fault?	[2]
f)	What is thrashing? How to avoid thrashing?	[3]
g)	What is a bit vector?	[2]
h)	Give an overview of directory structure.	[3]
i)	What are the goals of protection?	[2]
j)	What is language based protection?	[3]

PART - B

(50 Marks)

2.a) What are operating system objectives and functions?
b) What are the real-time features systems and time-shared systems? [5+5]

OR

3.a) What are batch systems? What are its advantages and disadvantages?
b) What is a system call? What are different types of system calls? [5+5]

4. Simulate FCFS, SJF and priority scheduling algorithms on following data and compute average waiting time: [10]

PID	ARRIVAL TIME	BURST TIME	PRIORITY
P01	0	15	2
P02	1	6	1(highest priority)
P03	4	9	5(lowest priority)
P04	5	2	3
P05	6	7	4

OR

5. What is process synchronization? Describe peterson's solution to critical section problem. [10]

6. Explain FIFO and LRU page replacement techniques with examples. [10]

OR

7. Discuss about segmentation with paging memory management technique. [10]

8. Explain LOOK and C-LOOK disk scheduling algorithms with example. [10]

OR

9. Explain about file system mounting and file protection. [10]

10. What are principles of protection? What do you mean by revocation of access rights? Why is it performed? [10]

OR

11. Explain banker's algorithm for deadlock avoidance and state its advantages and disadvantages. [10]

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